ASSIGNMENT No: 10 Title: Implement Page Rank Algorithm. (Use python or beautiful soup for implementation). Problem Statement: Implement Page Rank Algorithm. (Use python or beautiful soup for implementation).

Aditya Rade Roll.No: BEAD23167

Step 1: Import NetworkX

In [32]:
```
import networkx as nx
```

Step 2: Create the Graph

In [33]:
```
G = nx.barabasi_albert_graph(60, 41)

print("Number of Nodes:", G.number_of_nodes())
print("Number of Edges:", G.number_of_edges())
```

```
Number of Nodes: 60
Number of Edges: 779
```

Step 3: Calculate PageRank

In [34]:
```
pr = nx.pagerank(G, alpha=0.4)

print("PageRank Values:")
print(pr)
```

```
PageRank Values:
{0: 0.028034429310100223, 1: 0.012358273408648503, 2: 0.01338840428275777, 3: 0.0133
77263726068364, 4: 0.013380109265520076, 5: 0.012785440568089015, 6: 0.0127763996852
46973, 7: 0.013188903448347921, 8: 0.012760651908892053, 9: 0.012760778772323032, 1
0: 0.012358721149347513, 11: 0.01316229657641856, 12: 0.012962024780841463, 13: 0.01
3169393515676898, 14: 0.01257178788234387, 15: 0.012767003975230694, 16: 0.012551607
291431366, 17: 0.012594955264860994, 18: 0.01338381951934189, 19: 0.0131554681003591
2, 20: 0.013367325976135738, 21: 0.013167973304878705, 22: 0.013580201176648387, 23:
0.011981153688120237, 24: 0.01336360918811309, 25: 0.013159482014595677, 26: 0.01357
2008316086223, 27: 0.012574891369081551, 28: 0.013785060313582634, 29: 0.01297013793
7445052, 30: 0.012574667680632989, 31: 0.012178557282474403, 32: 0.0133634389253301
6, 33: 0.013572178578869153, 34: 0.012785082721746966, 35: 0.012354883239287167, 36:
0.013379552550898805, 37: 0.01336343892533016, 38: 0.01298929033004309, 39: 0.012782
113002520438, 40: 0.012781978916987164, 41: 0.01258745913306594, 42: 0.0276463025434
6684, 43: 0.027330873251784495, 44: 0.026837396264873026, 45: 0.026311826442882595,
46: 0.02588251064312399, 47: 0.025749081232591377, 48: 0.02533802235360387, 49: 0.02
5037793976807332, 50: 0.02473991579750088, 51: 0.02395653512682098, 52: 0.0236362695
93205687, 53: 0.02304516394512364, 54: 0.023212010967162924, 55: 0.02291089019931215
3, 56: 0.02290228138820977, 57: 0.0220218287910739, 58: 0.021899296721041597, 59: 0.
02181978375769502}
```

Step 4: Display PageRank in Descending Order

In [35]:
```
sorted_pr = sorted(pr.items(), key=lambda x: x[1], reverse=True)
```

```
print("PageRank in Descending Order:\n")

for node, score in sorted_pr:
    print(f"Node {node}: {score:.6f}")
```

```
PageRank in Descending Order:

Node 0: 0.028034
Node 42: 0.027646
Node 43: 0.027331
Node 44: 0.026837
Node 45: 0.026312
Node 46: 0.025883
Node 47: 0.025749
Node 48: 0.025338
Node 49: 0.025038
Node 50: 0.024740
Node 51: 0.023957
Node 52: 0.023636
Node 54: 0.023212
Node 53: 0.023045
Node 55: 0.022911
Node 56: 0.022902
Node 57: 0.022022
Node 58: 0.021899
Node 59: 0.021820
Node 28: 0.013785
Node 22: 0.013580
Node 33: 0.013572
Node 26: 0.013572
Node 2: 0.013388
Node 18: 0.013384
Node 4: 0.013380
Node 36: 0.013380
Node 3: 0.013377
Node 20: 0.013367
Node 24: 0.013364
Node 32: 0.013363
Node 37: 0.013363
Node 7: 0.013189
Node 13: 0.013169
Node 21: 0.013168
Node 11: 0.013162
Node 25: 0.013159
Node 19: 0.013155
Node 38: 0.012989
Node 29: 0.012970
Node 12: 0.012962
Node 5: 0.012785
Node 34: 0.012785
Node 39: 0.012782
Node 40: 0.012782
Node 6: 0.012776
Node 15: 0.012767
Node 9: 0.012761
Node 8: 0.012761
Node 17: 0.012595
Node 41: 0.012587
Node 27: 0.012575
Node 30: 0.012575
Node 14: 0.012572
```

```
Node 16: 0.012552
Node 10: 0.012359
Node 1: 0.012358
Node 35: 0.012355
Node 31: 0.012179
Node 23: 0.011981
```

Step 5: Find the Highest PageRank Node

In [36]:
```
highest_node = max(pr, key=pr.get)

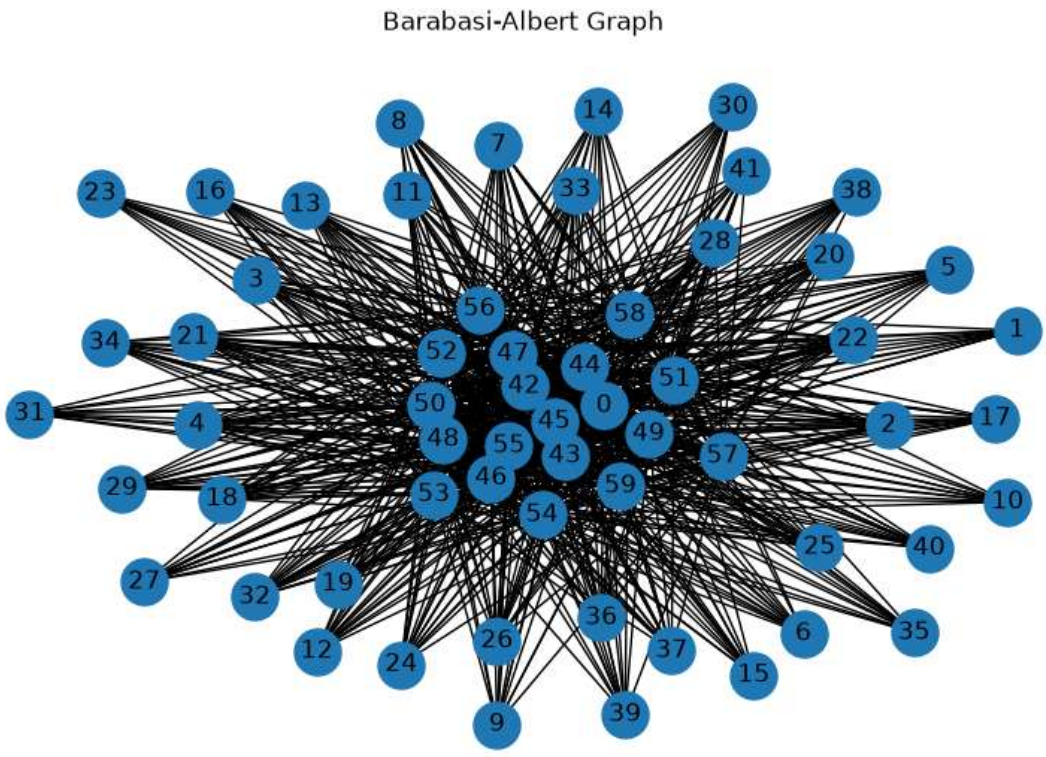
print("\nHighest PageRank Node:", highest_node)
print("Highest PageRank Score:", pr[highest_node])
```

```
Highest PageRank Node: 0
Highest PageRank Score: 0.028034429310100223
```

Step 6: Visualize the Graph

In [38]:
```
import matplotlib.pyplot as plt

plt.figure(figsize=(8, 5))
nx.draw(G, with_labels=True, node_size=500)
plt.title("Barabasi-Albert Graph")
plt.show()
print("Aditya Rade   BEAD23167")
```



Aditya Rade   BEAD23167